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Research paper

Performance of ancestry-informative SNP and microhaplotype markers

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ABSTRACT

The use of microhaplotypes (MHs) for ancestry inference has added to an increasing number of ancestry-informative markers (AIMs) for forensic application that includes autosomal single nucleotide polymorphisms (SNPs) and insertions/deletions (indels). This study compares bi-allelic and tri-allelic SNPs as well as MH markers for their ability to differentiate African, European, South Asian, East Asian, and American population groups from the 1000 Genomes Phase 3 database. A range of well-established metrics were applied to rank each marker according to the population differentiation potential they measured. These comprised: absolute allele frequency differences (δ); Rosenberg's informativeness for (ancestry) assignment (I_n); the fixation index (F_{ST}); and the effective number of alleles (A_e). A panel consisting of all three marker types resulted in the lowest mean divergence per population per individual (MDPI = 2.16%) when selected by I_n. However, when marker types were not mixed, MHs were the highest performing markers by most metrics (MDPI < 4%) for differentiation between the five continental populations.

1. Introduction

Sequencing of the entire human genome is becoming increasingly faster and more accessible with the development of next generation sequencing (NGS) technology. It has assisted the discovery of numerous trait-linked markers for disease diagnosis and forensic analyses [1,2]. The greater the number of markers used, the more accurate ancestry predictions are supposed to be [3]. However, this has not been proven and a refined panel of efficient markers are favoured for forensics, particularly in multiplex assays where small and robust panels maintain the all-important sensitivity of the assay, as well as reducing costs and use of limited evidential material.

To predict the biogeographical ancestry (BGA) of a donor requires ancestry-informative markers (AIMs) with maximum genetic differentiation between populations, ideally including those with one allele fixed in one or more populations and alternative alleles fixed in other populations. Yang et al. [4,5] ranked short tandem repeat (STR) markers by "marker efficiency" which we will describe as "differentiation potential". This was based on a measure which took half the sum of the absolute allele frequency differences between them. It focused on two populations commonly assumed to be admixed: European Americans and African Americans. Ding et al. [6] investigated the potential of autosomal single nucleotide polymorphisms (SNPs) to differentiate between Africans and Europeans and between Chinese and Japanese by various measures.

Different selection metrics have been applied to rank and filter candidate sets of highly informative ancestry loci to maximise differentiation between populations [6,7]. Most common metrics include allele frequency differences (δ), Rosenberg's informativeness for assignment (I_n), and the fixation index (F_{ST}). These metrics have been used to select amongst autosomal SNPs and microhaplotypes (MHs) [8–10]. In studies that compared panel performance using SNP data, I_n and F_{ST} were highly concordant and amongst the most accurate for ancestry assignment [6,11,12].

The estimation of F_{ST} has been defined differently in various studies with varied assumptions and parameter interpretations applied [13–15]. F_{ST} was originally introduced by Wright [16,17] as a ratio of variances between bi-allelic loci based on allele frequency differences between subpopulations. However, this assumes there are no fluctuations in allele frequency in the population, which is possible if population sizes are large, while mutation and selection pressures are absent. Wright's F_{ST} is bounded by zero and one, and can lead to overestimation

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of the genetic difference between populations in small sample sizes [18]. Kimura [19] later extended the scenario of random genetic drift between bi-allelic loci to multiple alleles. This is incorporated in Nei's adaptation of F_{ST}, a generalisation of Wright's metric that produces the same result in the case of two alleles [20].

Small negative F_{ST} values have generally been interpreted as artefacts of sampling and rounded to zero. However, when genetic differentiation is very close to zero or when there is greater heterozygosity within a sub-population than in the total population (*i.e.* an apparent violation of the Wahlund effect at any locus [21]), F_{ST} estimators can produce negative values at individual loci. It should be remembered that F_{ST} has historically been used to measure genetic distance between populations and not for the selection of AIMs. It has often been averaged over multiple loci, and in such cases occasional negative values are countered by positive values.

More recently, a measure of allele diversity has been introduced which draws upon the effective number of alleles, A_e [22]. Markers with equally frequent alleles are scored highly and rank well for their ability to detect mixtures. This has contributed to the selection of MHs with dual ability for mixture deconvolution and ancestry inference [9,22,23].

Uni-parentally inherited markers trace a single lineage, which is highly informative for inferring ancient ancestral membership, but can be misleading for the inference of more recent ancestry, particularly when there is population admixture. Autosomal markers, on the other hand, are inherited from both parents and can therefore reveal more recent co-ancestry contributions. Ancestry-informative SNPs are favoured over STRs because of their low mutation rates, however SNPs are restricted to mostly two, or more rarely, three and four alleles (bi-, tri- and tetra-allelic loci), which limits the number of unique populations that can be differentiated by any one marker. This also makes SNPs poor identifiers of mixtures whereas the highly variable repeat regions of STRs provide better discrimination between individual DNA sources. MHs are a combination of multiple SNP loci within 300 base pairs (bps) and show promise in providing the same ancestral information as individual SNPs, with the benefit of low mutation rates, while also detecting higher levels of variability from multi-allelic markers [22]. MHs therefore have the potential to supplement ancestry-informative SNP panels with their multiple applications which include, but are not limited to, ancestry inference, mixture detection, and human identification. They could potentially be applied to missing persons identification, relationship testing, and medical diagnostics [23–26].

This study focuses on the selection of AIMs for population inference at the continental level that have now become amenable to NGS analysis. Different population differentiation metrics were applied to explore three types of markers: bi-allelic SNPs, tri-allelic SNPs, MHs, and a combination of all three to determine which metric provides the best separation amongst reference populations from the selection of the top 30 AIMs. Markers were analysed for five major continental populations: African (AFR), European (EUR), South Asian (SAS), East Asian (EAS), and indigenous (or native) American (AME). To avoid selection of an imbalanced AIM panel, over-populated with markers specific for a single population, we propose a method for sequential addition of AIMs to a forensic panel based on the cumulative differentiation potential that builds for each reference population. With this approach, we determine whether multiplexes of mixed markers are preferable over those using one marker type.

2. Materials and methods

2.1. Candidate AIMs

Bi-allelic SNP markers were selected from Kiddlab's 55 AIMs [27], Seldinlab's 128 AIMs [28], and the EUROFORGEN Global AIM-SNP set [10]. A total of 36 tri-allelic SNPs with contrasting allele frequency proportions amongst populations were selected from 1000 Genomes Phase 3 [29]. They included the six tri-allelic loci in the Global AIM-SNP set (rs2069945, rs2184030, rs433342, rs4540055, rs5030240, and rs17287498) [10], two of which (rs4540055 and rs5030240) are also part of the SNP*for*ID 34-plex ancestry panel [30]. MHs were sourced from Kiddlab's 130 MH selection [9] excluding two loci (mh11KK-091 and mh07KK-081) that became single bi-allelic SNPs after removal of their insertion-deletion polymorphisms (indels rs1799732 and rs41303343, respectively). Altogether, our analysis of the full range of current viable forensic AIMs consisted of 272 bi-allelic SNPs, 36 tri-allelic SNPs, and 128 MHs. Genotypes for these loci in five populations (AFR, EUR, SAS, EAS, AME) were then compiled from the 1000 Genomes Phase 3 variant database (Supplementary File S1).

2.2. Metrics for ranking ancestry-informative markers

Population-specific differentiation potential was calculated based on four different metrics: absolute allele frequency difference (δ), Rosenberg's informativeness (I_n), fixation index (F_{ST}), and effective number of alleles (A_e). These were calculated for pairwise population groupings (*e.g.* AFR *versus* non-AFR, EUR *versus* non-EUR, *etc*). Let $p_{i,j,k}$ be the relative frequency of the ith allele at the jth locus in the kth population (AFR, EUR, SAS, EAS or AME). Then let $q_{i,j,k}$ be the relative frequency of the same allele at the same locus in all other populations combined. We then define δ as the summed absolute allele frequency differences [5,31]:

$$\delta_{j,k} = \sum_i |p_{i,j,k} - q_{i,j,k}| \tag{1}$$

We considered other permutations of δ, including the maximum absolute allele frequency difference across all alleles (δ_{max}) and the average absolute allele frequency difference across all alleles (δ_{ave}), given by:

$$\delta_{j,k}^{max} = \max_i |p_{i,j,k} - q_{i,j,k}| \tag{2}$$

$$\delta_{j,k}^{ave} = \text{average}_i |p_{i,j,k} - q_{i,j,k}| \tag{3}$$

Rosenberg's informativeness for assignment [32] with K populations is defined by:

$$I_n = \sum_i \left(-p_{i,j} \ln p_{i,j} + \frac{1}{K} \sum_k (p_{i,j,k} \ln p_{i,j,k}) \right) \tag{4}$$

where $p_{i,j}$ is the frequency of the ith allele at the jth locus in all populations combined. For our pairwise population comparisons, this reduces to:

$$I_{j,k} = \sum_i \left(-p_{i,j} \ln p_{i,j} + \frac{1}{2} (p_{i,j,k} \ln p_{i,j,k} + q_{i,j,k} \ln q_{i,j,k}) \right) \tag{5}$$

The F_{ST} calculation in this study is adapted for multiple alleles similar to Nei [20,33] where:

$$F_{j,k} = \frac{H_T - H_S}{H_T} \tag{6}$$

H_T is the heterozygosity in all populations combined:

$$H_T = 1 - \sum_i p_{i,j}^2 \tag{7}$$

and H_S is the average heterozygosity over all populations:

$$H_S = \frac{1}{K} \sum_k \left(1 - \sum_i p_{i,j,k}^2 \right) \tag{8}$$

For our pairwise population comparisons, this reduces to:

$$H_S = \frac{1}{2}\left(\left(1 - \sum_i p_{i,j,k}^2\right) + \left(1 - \sum_i q_{i,j,k}^2\right)\right) = 1 - \frac{1}{2}\left(\sum_i (p_{i,j,k}^2 + q_{i,j,k}^2)\right) \quad (9)$$

For these metrics, the minimum differentiation potential of zero occurs when there is no genetic structuring across all populations, and allele frequencies are equal in all populations [32]. The maximum differentiation potential results when there is a unique, fixed allele in each population. The A_e metric, however, is at a maximum for markers with equally frequent alleles, and draws upon the Simpson index of diversity [34–36]. It has been proposed as a metric to select for the ability to detect mixtures and is calculated as the reciprocal of expected homozygosity [22,34] according to:

$$A_{j,k} = \frac{1}{\sum_i p_{i,j,k}^2} \quad (10)$$

We include it to test the hypothesis that A_e can be a metric for both differentiation potential and mixture deconvolution. As AIMs benefit from fixed alleles (*i.e.* alleles with close to 100% frequency in a single population, with large allele frequency differences between populations) we ranked AIMs in two ways for this metric: from highest value to lowest value (A_e) and the reverse ranking (-A_e) in order to assess the effect on differentiation potential.

2.3. Negative values for differentiation potential

In this study, δ and A_e are the only metrics restricted to positive output values (Eqs. 1–3 and 10). The literature remains ambiguous on how to treat negative I_n and F_{ST} values when selecting AIMs. Negative values tend to occur when there are apparent violations of the Wahlund effect (*i.e.* $H_S > H_T$) for a particular locus. It is possible for a marker to be heterozygous in one population and homozygous for the same single allele in all others, rather than the other way around. Such markers still have good differentiation potential, even though the pattern of variation will produce a negative value for I_n and F_{ST}. It is only zero values that indicate poor differentiation potential.

Negative F_{ST} values have often been rounded to zero but this represents an under-estimation of the marker's differentiation potential for $F_{ST} < < 0$. F_{ST} in fact ranges from -∞ to 1, an asymmetric distribution around zero. Taking absolute values will bias the selection towards markers with large negative F_{ST}. Ideally, the two ranges for F_{ST} ($F_{ST} < 0$ and $F_{ST} > 0$) should be assessed separately; although this then divides the marker set into two. We have thus selected markers with negative F_{ST} if the absolute value was greater than the marker with the maximum positive F_{ST}. To avoid large jumps in the cumulative differentiation potential for $F_{ST} < < 0$, all F_{ST} values less than -1 were rounded up to -1 to restrict the range between [-1, 1].

In contrast, I_n, is symmetrical with no range restrictions. The differentiation potential for a selected marker is indicated by the magnitude of I_n, not its sign. Therefore, the absolute value of I_n was taken in this study.

2.4. AIM selection process

Markers were sorted into four panels: (i) bi-allelic SNPs only; (ii) tri-allelic SNPs only; (iii) MHs only; and (iv) bi-allelic SNPs, tri-allelic SNPs, and MHs together (termed: all-marker panel). For each marker within each of the four panels and for each of the five populations, a pairwise differentiation potential was calculated using one of the four metrics in turn and ranked from highest potential to lowest potential (Supplementary File S1). The A_e metric was additionally ranked in reverse, as described previously. Some populations have higher differentiation potentials for most metrics. For example, AFR is the most easily differentiated of the five populations and has many markers with higher potential than the most highly differentiating markers in all other populations. As a result, it requires less markers in a balanced panel to achieve the same cumulative differentiation potential as other populations. A balanced panel will have sufficient markers to differentiate each population (and no more) so that the least well-differentiated populations will require more markers than others.

To avoid unbalanced panels, dominated by single population-informative markers, a stepwise selection process was applied. For each reference population, every marker was ranked from most discriminating (highest potential) to least discriminating (lowest potential) based on their population-specific differentiation potential (with A_e also ranked in reverse). The first marker selected had the lowest maximum potential over all reference populations. Consecutive markers were then chosen from the population with the lowest cumulative potential in the selected panel such that the cumulative potential for all populations was raised together. Markers were chosen from the top of the ranked lists and removed from each population after selection to avoid duplication. The cumulative differentiation potential was then updated for each population in the selected panel and the next marker to be chosen was from the population with the lowest cumulative sum.

This approach is similar to the method proposed by Zhao et al. where four steps were described for the partitioning, filtering, selection, and optimisation of markers [37]. While Zhao's study used F_{ST} for a pre-filtering step before applying a feature selection algorithm to obtain a SNP panel that maximised classification between populations; our study employs a range of metrics, including F_{ST}, to calculate the differentiation potential for each marker, upon which selection is directly based.

The calculation of each metric was coded in a custom R script (Supplementary File S2) and run in RStudio with R version 3.4.2 [38]. Custom candidate marker sets can be imported and each marker ranked in order of differentiation potential. A balanced AIM panel of m markers for global differentiation is then returned, where m is chosen by the user. Because the number of candidate tri-allelic SNPs was the lowest of any marker type ($n = 36$), we set $m = 30$.

2.5. Panel assessment and population differentiation

After selection of the top 30 AIMs using each metric, population differentiation was assessed using the admixture model in Pritchard's STRUCTURE program [39,40] with correlated allele frequencies [40]. We ran 10,000 burn-ins and 10,000 MCMC replications for 30 iterations at $K = 5$. This number of iterations was sufficient for the major and minor modes to be identified by the online CLUMPAK pipeline [41] for each simulation. For each individual, any proportion of non-self-declared ancestry was averaged across the number of reference populations so that an individual who, for example, self-declared to be African and was assigned 100% African by STRUCTURE would have a zero non-self-declared ancestry proportion. The mean non-self-declared ancestry proportion (also termed the mean divergence per population per individual, MDPI [42]) was then calculated to compare classifier output.

3. Results

A set of candidate markers for ancestry inference amongst five continental populations were ranked based on δ, I_n, F_{ST}, and A_e estimates. STRUCTURE plots and cumulative differentiation potentials are shown in Supplementary File S3. Individual plots are presented for every combination of panel and differentiation potential metric, together with the number of repetitions comprising the major mode (of a total of 30) and the MDPI values.

The comparison between markers and metrics highlights the success of I_n as an AIM selector, as well as the superior performance of MH panels over bi-allelic SNPs to differentiate populations when selected with δ, F_{ST}, or A_e (Table 1). Despite having different MDPI values, I_n and F_{ST} were well correlated with a Spearman's coefficient $\rho > 0.6$ for all panels (Supplementary File S4) and 21 of the 30 selected markers in common for the all-marker panel (Fig. 1). Pairwise significance

Table 1
The mean divergence (proportion of non-self-declared ancestry) per population per individual (MDPI) from panels consisting of the top 30 ancestry-informative markers as ranked by different selection metrics.

Marker panel	δ	I_n	F_{ST}	A_e	$-A_e$
bi-allelic	5.13%	2.17%	7.02%	3.67%	4.48%
tri-allelic	8.50%	8.57%	8.74%	9.13%	8.90%
MH	2.97%	2.70%	3.64%	3.08%	14.30%
all	2.20%	2.16%	7.02%	3.15%	3.63%

Table 2
Numbers of each marker type in the top 30 ancestry-informative markers in the all-marker panel using different selection metrics. The pool of candidate markers included bi-allelic SNPs, tri-allelic SNPs, and microhaplotypes.

Metric	Bi-allelic SNP	Tri-allelic SNP	Microhaplotype
δ	22	4	4
I_n	22	1	7
F_{ST}	30	0	0
A_e	0	1	29
$-A_e$	26	2	2

comparisons between all tests can be found in Supplementary File S4.

The most accurate BGA predictions reporting the lowest MDPI of 2.16% was the all-marker panel selected with I_n (Table 1). This panel was highly correlated with the bi-allelic panel ($\rho = 0.83$), which had the second lowest MDPI of 2.17%. Of the five metrics tested (including $-A_e$), Rosenberg's I_n shared the most markers with all other metrics. The least correlated was the A_e metric with 27 unique markers (Fig. 1).

Panels consisting of only tri-allelic markers were the worst performing for all metrics, except $-A_e$. The tri-allelic markers in this study showed limited ability to differentiate between populations and were consistently unable to differentiate between EAS and AME. This difficulty in accurately assigning BGA to AME individuals has been noted elsewhere [43].

Given that A_e favours equally frequent alleles, which are more likely to occur between the multiple allelic variants of MH markers than SNPs, A_e was the only metric that selected more MH markers than SNPs in the all-marker panel (Table 2). This improved the panel performance compared to its selection of the bi-allelic SNPs panel (Table 1). Although use of F_{ST} resulted in an all-marker panel consisting of only bi-allelic SNPs (Table 2), it was the MH panel that resulted in the lowest MDPI (Table 1).

Overall, MHs were the better performing panel, as indicated by most metrics (excluding $-A_e$), when marker types were not mixed. The only single marker panel that performed better than MHs was the bi-allelic SNP panel selected by Rosenberg's I_n, but even then there was less than 1% difference in MDPI (Table 1).

4. Discussion

Ancestry prediction from genotyping well-chosen marker panels has been possible with the discovery of numerous AIMs in the human genome. Smaller panel sizes are desirable to increase forensic efficiency and lower costs. However, the number of candidate markers available for ancestry has far exceeded the minimum number required for successful distinction between continental populations. In this study we compared various selection metrics to rank and filter a large collection of AIMs. Since the original publication of many SNP markers [27,28,30,44,45], MHs have more recently been advocated as superior for population differentiation [8,9,24]. These haplotype blocks are strings of SNPs within 300 bp segment sizes or less, and replicate the high variability of multi-allelic loci, which is ideal for mixture detection.

Our comparison shows that some metrics tend to over-estimate the differentiation potential of bi-allelic SNPs when all marker types are combined. Despite δ and F_{ST} selecting an all-marker panel consisting predominantly of bi-allelic SNPs (> 70%), the MH only panels had a lower MDPI (*i.e.* were better performing) than the bi-allelic SNP panels (Table 1). These two metrics are distance measures derived from differences between allele frequencies (δ) and heterozygosities (H). They will bias marker selection towards large population differences between allele frequencies. This is more common for bi-allelic SNPs, which can be close to fixation for one allele in one population and for the alternative allele in another population. MHs have more variants, so the heterozygosity in all populations combined (H_T) and in sub-populations (H_S) tend to be larger. Because F_{ST} decreases as H_T and H_S increase (Eq. 6), MHs are ranked lower in differentiation potential than most bi-allelic SNPs and as a result, MHs were not amongst the top 30 markers to be selected for the all-marker panel using F_{ST}. The use of I_n and A_e, on the other hand, resulted in the most MHs in the all-marker panels (Table 2) and similar (low) MDPIs for MH and all-marker panels

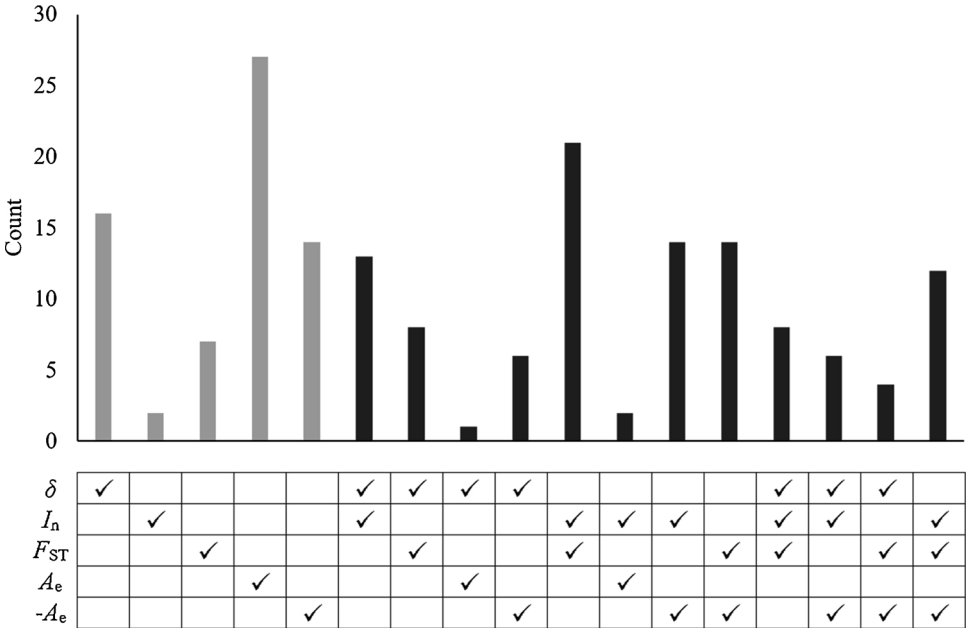


Fig. 1. The number of non-overlapping (grey) and overlapping (black) ancestry-informative markers (AIMs) in the all-marker panel from the top 30 for each metric and combinations of metrics. Markers were selected from bi-allelic SNPs, tri-allelic SNPs, and microhaplotypes. A tick (✓) indicates selection with the corresponding metric (rows) for any particular combination of metrics (columns). For example, the last column indicates that there were 12 overlapping AIMs in the top 30 for the all-marker panels selected with I_n, F_{ST}, and $-A_e$.

(Table 1). This is because I_n is based on Shannon's information measure or entropy ($plnp$) and, together with A_e, is a measure of allelic diversity with more alleles resulting in a higher index [34,36].

The worst performing panels were those that consisted of only tri-allelic SNPs. Although capable of differentiation between AFR, EUR, and EAS, they could not separate AME from EAS when selected by any metric, as indicated by STRUCTURE analysis at $K = 5$ (Supplementary File S3). Previous studies have also reported difficulty in the genetic differentiation of EAS and AME at lower K values, but the two populations are often separated with bi-allelic SNPs, indels, or MHs [28,43,44,46,47]. The EAS-AME similarities are likely to be at least partly explained by the common ancestors of indigenous Americans and East Asians tracing back to a northeast Asian gene pool near Siberia about 40,000 years ago, which is relatively recent compared to the common gene pools shared by other continental populations [48–50].

Less than 14% of selected markers in the all-marker panels were tri-allelic (Table 2), none of which have previously been included in the SNP*for*ID 34-plex [30] or EUROFORGEN's global AIM-SNP set [10]. We suspect this is because these panels were not created to differentiate SAS. Regardless of this finding, even for the present study, variation within SAS was greater than variation between EAS and AME for δ, I_n and F_{ST} (Supplementary File S3) with the regular appearance of a genetic cline. Principal component analysis (PCA) plots of Central/South Asian populations have shown them to be intermediate between EUR and EAS [44,45] and are likely to consist of multiple ancestral sources [51]. We conclude that tri-allelic SNPs are not sufficient for global differentiation without the aid of complementary AIMs with different patterns of variation.

The five continental reference populations were generally well differentiated by each metric and for each marker panel, but the most consistent and the best performing sets, regardless of the metric used, were the all-marker panels (Table 1). The most accurate metric and marker combination was an all-marker panel selected with I_n. Markers selected with δ and I_n resulted in an improvement in panel performance when bi-allelic, tri-allelic, and MHs were combined, whereas for F_{ST} and A_e, the MH only panel was the best.

4.1. Allele frequency differences as a measure of differentiation potential

Since large allele frequency differences between populations are generally good indicators of informative AIMs, the differentiation potential for each marker could simply be represented by the maximum allele frequency difference per marker (δ_{max}). However, with this approach it is possible that a bi-allelic marker and a multi-allelic marker with more than two alleles have the same δ_{max}, despite the fact that the multi-allelic marker has additional alleles with which to differentiate other populations (Table 3). In this case, δ_{max} is susceptible to overlooking multi-allelic markers with minor alleles unique to a single population.

Also considered was δ averaged over all alleles (δ_{ave}) to adjust for the varying number of alleles across different multi-allelic markers. In the scenario where a bi-allelic marker is fixed for one allele in *Population 1* and fixed for the alternative allele in *Population 2*, $\delta_{ave} = (1 + 1) / 2 = 1$. Now, if this same scenario was to occur for a tri-allelic marker where the third allele has a frequency of zero for both populations, $\delta_{ave} = (1 + 1 + 0) / 3 = 0.667$. Despite both the bi-allelic and tri-allelic markers being equally informative for two populations, the calculation of δ_{ave} is biased towards the marker with fewer allelic variants. As a result, we excluded δ_{max} and δ_{ave} from our comparisons.

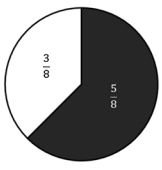
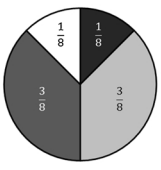
4.2. Effective number of alleles

A_e is an alternative selection method used to rank MHs based on the effective number of equally frequent alleles. Its principle application has been to rank MHs for mixture deconvolution [22,34]. Mixtures are best detected when allelic variation is high, a feature satisfied by MHs as opposed to bi- or tri-allelic SNPs which are restricted to a maximum of six genotype combinations. Markers with high heterozygosity (low homozygosity) will have a high A_e. However, markers with single, unique alleles fixed in different reference populations are favoured for ancestry inference.

Ranking by A_e was therefore reversed ($-A_e$) to select for alleles with high homozygosity. However, because A_e is simply the reciprocal of the summed square of the allele frequencies (Eq. 10), this metric does not account for pairwise population comparisons. Where a marker is homozygous for one allele in one population, it will not necessarily be homozygous for the alternative allele in another population.

A_e favours MH markers with high haplotype diversity. This was generally the case for top ranked markers by A_e. Despite similar haplotype frequencies between pairwise populations, there was usually at least one distinct allele that differentiated one or more reference populations. Markers with low A_e values (selection based on $-A_e$) only performed better than markers with high A_e values for the tri-allelic panel (Table 1). MHs that were poor for mixture deconvolution were also poor for ancestry inference, resulting in the worst MDPI of 14.30% (Table 1). Conversely, high performing ancestry-informative MHs were

Table 3
Example distribution of allele frequencies in two populations for a bi-allelic (A) and tetra-allelic (B) marker as well as δ values for each marker.

Marker	Population 1	Population 2	δ	δ_{max}	δ_{ave}
A			1	0.5	0.5
B			1.5	0.5	0.375

also good for mixture deconvolution, as found by Kidd et al. [9], the panel from which our candidate MH markers were sourced. However, better still for population differentiation were MHs selected using either δ or I_n. This fortuitous coincidence means that any small loss in differentiation potential resulting from the inclusion of MHs will be largely compensated by their improved ability to detect mixtures.

4.3. Balancing differentiation potential between populations

The selection procedure we applied set the requirement to balance the differentiation potential between populations and avoid an abundance of markers that differentiate AFR (for example) at the expense of markers more informative for other populations. Markers for difficult-to-differentiate populations were added to make the cumulative differentiation potential equivalent for all populations. This was more effective than if the top ranked markers were chosen without regard for differentiation potential across all populations. The latter approach is similar to that utilised in a study by Toma et al. [52] where an accuracy rate of 96.75% was achieved with 206 SNPs. Applying the requirement for balance in our selection procedure meant we obtained the same level of accuracy with only 30 markers.

AFR, the most genetically distinct of the five continental reference populations, is derived from a continent where anatomically modern humans have existed for over 200,000 years and from which the more recent populations of EUR, SAS, EAS, and AME were derived less than 100,000 years ago [53,54]. As expected, the AFR population was well differentiated with δ, I_n, and F_{ST}. Despite the preference for difficult-to-differentiate populations in our marker selection strategy, most markers still retained high AFR-specific differentiation potential and thus, the cumulative differentiation potential for AFR tended to exceed all other populations (Supplementary File S3).

Phillips et al. [10] highlighted that the calculation of absolute allele frequency difference (δ) focuses on pairwise comparisons between populations. With the distinct genetic composition of Africans, the difference in our pairwise comparisons will be larger for AFR *versus* non-AFR than for any other pairwise grouping. When AFR is grouped together in a "non-population" (*e.g.* non-EUR, non-EAS), its relatively large genetic distance will raise the within group variation. This is potentially why the AFR-specific differentiation potential for markers selected using δ tended to be very high (except for tri-allelic SNPs), with cumulative differentiation potential exceeding all other populations (Supplementary File S3). It also explains why bi-allelic SNPs selected using δ could not fully differentiate EUR and SAS. Conversely, the AFR-specific differentiation potential was not so dominant for markers selected with I_n and F_{ST}.

Various approaches were taken for the calculation of differentiation potential based on F_{ST} (data not shown). Due to the asymmetry of Nei's F_{ST}, inclusion of all markers ranked by absolute value tended to overestimate the differentiation potential of markers when $F_{ST} < 0$. When only large negative F_{ST} values were included without range restriction, this created large jumps in the cumulative differentiation potential, particularly for difficult-to-differentiate populations (*e.g.* AME). If only positive F_{ST} values were taken, this avoided the large step changes in cumulative differentiation potential but ran the risk of ignoring more suitable AIMs. These variations of F_{ST} resulted in similar MDPI values (within 1%) for corresponding marker panels (data not shown).

5. Conclusions

Four metrics for population differentiation potential were applied in this study to rank and compare the panel performance of both SNP and MH markers for five continental populations. Marker selection using Rosenberg's I_n metric produced the most accurate ancestry-informative marker panels. Despite some discrepancies in marker rankings between metrics, I_n and F_{ST} had the greatest overlap of markers (Fig. 1). This is consistent with the comparison by Ding et al. reporting the highest correlation between these two metrics [6].

As Kidd et al. [9] suggest, the increased allelic variability of MHs provides multiple benefits for future forensic multiplexes, including ancestry inference and mixture deconvolution, a limitation of bi-allelic SNPs due to their binary nature. Tri-allelic and tetra-allelic SNPs have also demonstrated greater potential for detecting multiple sources in a DNA sample as they have greater allelic variability than bi-allelic SNPs [55,56]. A_e has been published as a metric for selection of MH markers to maximise mixture deconvolution ability, many of which were coincidentally also good for inferring ancestry [22,23,26].

Overall, multiplexes consisting of both bi-allelic SNPs and MH markers consistently perform well and are accurate for distinguishing the five continental populations. However, when comparing individual marker types, our results show that MHs outperform bi-allelic and tri-allelic SNPs for three of the four metrics (δ, F_{ST}, and A_e). Where they do not perform better than SNPs (*e.g.* I_n), the loss in performance is minimal.

Finally, we demonstrate that a panel with as few as 30 well-chosen AIMs can differentiate five continental level populations. This is possible when AIMs are added sequentially so as to build and maintain a balanced cumulative differentiation potential between each population, resulting in more AIMs for poorly differentiated populations and less for easily differentiated populations.

Declaration of Competing Interest

The authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.fsigen.2019.102141.

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